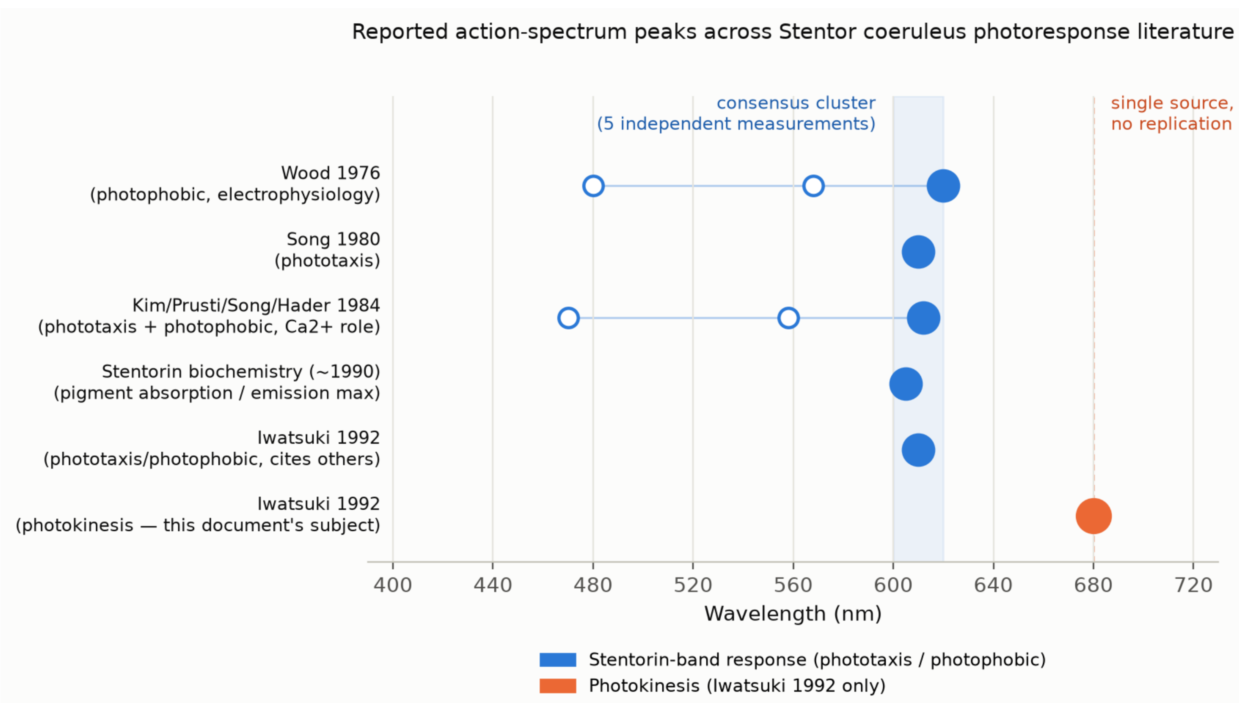
Stentor coeruleus Wavelength Selection Guide

Purpose: pool everything we've read across the *S. coeruleus* photoresponse literature into one decision — which wavelength(s) should StentorCam experiments prioritize to get a positive result. This is the decision document; for general background and citation-heavy detail (including a figure-by-figure critique of Iwatsuki 1992's axis units and 0.58 mm/s criterion) see `PHOTOKINESIS_ACTION_SPECTRUM_NOTES.md`. For the full experimental protocol once a wavelength is chosen (chamber design, controls, statistics, plate layout), see `ROBOCAM_ASSAY_PROTOCOL.md`, adapted from a design document our PI provided — it independently reaches the same 610/620 nm recommendation and treats 680 nm as a control wavelength, corroborating this guide.

Recommendation

- **Test 610-620 nm first.** Five independent measurements across three labs and 16 years (1976-1992) converge here — the strongest, most replicated signal in the whole literature.
- **Backup/secondary candidates: ~560 nm and ~470-480 nm.** Now corroborated by two independent papers (Wood 1976: 568/480 nm; Kim/Prusti/Song/Hader 1984: ~555-560/470 nm) — the same two-secondary-peak shape shows up twice, which is reassuring, though still lower-confidence than the 610-620 nm primary peak. Wood's 568 nm point was never directly tested in his own action spectrum (only inferred from absorption data); Kim/Prusti/Song/Hader's ~555-560 nm point was directly measured.
- **Deprioritize 680 nm.** This is Iwatsuki's claimed photokinesis peak, and it's an outlier by every measure we could check: single source, never independently replicated, no known matching pigment, and it's consistent with our own data showing little response there. Keep it as an exploratory/comparison arm, not the main bet.
- **Novel angle worth pursuing regardless of the above:** none of the five 610-620 nm measurements were photokinesis experiments — they were phototaxis or photophobic (avoidance) response. Testing 610-620 nm specifically for *photokinesis* is a measurement nobody has published. A positive result there would suggest photokinesis shares stentorin's photoreceptor rather than requiring Iwatsuki's proposed second, unidentified 680 nm photoreceptor.

Cross-paper synthesis



Source	Method	Response type	Reported peak(s)
Wood 1976	Electrophysiology + ciliary-reversal threshold	Photophobic	620 nm primary; 480 nm shoulder; 568 nm secondary (absorption data only, not directly tested)
Song 1980	Fluence-response curves, focused-beam orientation	Phototaxis	~610 nm (broad maximum)
Kim/Prusti/Song/Hader 1984	Ca2+-flux / fluence-response, quantum-efficiency action spectrum vs. absorption spectrum	Phototaxis + photophobic	~610-615 nm primary; ~555-560 nm secondary; ~470 nm shoulder
Stentorin biochemistry (~1990, Song lab)	Absorption / fluorescence-emission spectroscopy of the isolated pigment	Pigment characterization	~600-620 nm
Iwatsuki 1992	Fluence-response curves (cites the above)	Phototaxis/photophobic	610 nm
Iwatsuki 1992	Fluence-response curves, criterion velocity	**Photokinesis**	**680 nm — the outlier**

Wood 1976, Song 1980, and Kim/Prusti/Song/Hader 1984 values above are confirmed directly from the primary-source PDFs (now in `references/`) — the ILL request for Kim/Prusti/Song/Hader came through. Its Fig. 1 plots a proper quantum-efficiency action spectrum directly against stentorin's measured absorption spectrum (in vivo and at 77 K), and its multi-peak shape (~610-615 nm primary, ~555-560 nm secondary, ~470 nm shoulder) independently reproduces Wood 1976's pattern (620/568/480 nm) almost exactly — two separate labs, two different methods (fluence-rate criterion-response vs. direct quantum-efficiency calculation), converging on the same multi-band shape. That's a much stronger corroboration of the 610-620 nm cluster than a single-paper result. Only the stentorin-biochemistry summary is still from a secondary source pending full-text access.

The photoreceptor behind the 610-620 nm cluster is identified: **stentorin**, a hypericin-based chromophore with absorption confined to roughly 525-630 nm. It has essentially no absorption at 680 nm — which is exactly why Iwatsuki had to propose a second, distinct photoreceptor system to explain the photokinesis result. No follow-up work in the 30+ years since has identified what that second photoreceptor might be, and *S. coeruleus* has no plastids/chlorophyll that would otherwise explain red-region sensitivity out at 680 nm.

Why 680 nm specifically looks weak (summary)

- **Single source.** Of 24+ papers citing Iwatsuki (1992), none independently re-measured the photokinesis action spectrum — all just cite it as background.
- **No replication attempt succeeded.** Kühnel-Kratz & Häder (1994) is the one paper that re-examined photokinesis behavior with improved 3D tracking, but they used white light only — no monochromatic wavelength sweep — so it neither confirms nor refutes 680 nm.
- **No matching pigment.** See above.
- **Thin derivation of the criterion response.** Iwatsuki's action spectrum is built by finding the fluence rate needed to hit a fixed criterion velocity (0.58 mm/s) at each wavelength; the paper doesn't explain why 0.58 mm/s specifically, beyond it being a value within the overlapping range of all nine tested curves. (Full breakdown in `PHOTOKINESIS_ACTION_SPECTRUM_NOTES.md`.)

Other experimental variables to control for

- **Fluence-rate ceiling.** Kühnel-Kratz & Häder (1994) only saw positive photokinesis below ~10 W/m^2; above that, no clear kinesis — in tension with Iwatsuki's claim of an almost-linear velocity-vs-log-fluence relationship across the *entire* tested range. Check where our fluence rates fall relative to this cutoff.
- **Chamber geometry.** Kühnel-Kratz & Häder's dark-swimming baseline velocities (1.5-1.8 mm/s) were notably higher than every previously published value (0.2-1.2 mm/s, including Iwatsuki's), attributed to their wider 25 mm cuvette reducing wall effects. Our own baseline velocities may not be directly comparable to the literature unless our chamber width/depth is accounted for.
- **Light-source spectral purity.** Confirm actual output spectrum at whichever wavelength setting we test (LED/filter bandwidth can shift or wash out a narrow real peak) — relevant whether we're testing 610-620 nm or 680 nm.

Light sources: what the papers used vs. our apparatus

What the classic papers actually used (confirmed from the primary-source PDFs, now all in `references/`) — none used lasers or LEDs; all were filtered lamp/projector setups:

Paper	Light source	Wavelength selection
Iwatsuki 1992	Incandescent lamp, IR cut-off filter + water container	Interference filter (<6 nm half-bandwidth) + neutral-density filters; EG&G photometer
Wood 1976	Zircon arc lamp (behavioral test); cool white fluorescent bulbs (electrophysiology test)	Interference filters, 10 nm half-bandwidth, stepped every 20 nm 480-660 nm; ND + heat filters
Song 1980	40 W incandescent lamp, collimated	ND filters (white light); interference filters (Baird Atomic, ~10 nm half-bandwidth); Kettering Model 65 radiometer
Kim/Prusti/Song/Hader 1984	300 W slide projector (Gold)	Interference filters, ~10 nm half-bandwidth; Kettering Y.S.I. Model 65A radiant power meter
Kuhnel-Kratz & Hader 1994	15 W Zeiss microscope lamp (viewing); 150 W halogen (Schott KL 1500 / Osram Xenophot, stimulus)	White light only — no wavelength filtering, consistent with them never testing an action spectrum

Our apparatus: the plotting scripts (`full_data_plot.py`, `track_plot.py`) label the stimulus "LED On"/"LED Off" in their legends, even though the README's CLI flags are loosely named after "laser color" — that's historical naming, not a hardware spec, and no BOM/part number for either an LED or a laser diode exists in this repo (hardware specifics live in the separate RoboCam repo).

Current build direction: a laser-based coaxial illuminator with a 50/50 beamsplitter, sharing the optical axis with the camera. For that specific design, a laser diode is the better fit optically (small collimated beam couples through a beamsplitter far more efficiently than an LED, which needs extra collimating optics and loses more power to etendue mismatch). Recommended source: a **~605-620 nm orange laser diode module** — this wavelength sits in what's called the semiconductor "yellow-orange gap" (570-625 nm is hard for direct diode lasers), so it's a specialty part rather than an off-the-shelf pointer diode, but real products exist (RPMC Lasers' 590-619 nm orange category, Laserland's 600-640 nm modules, Thorlabs' visible laser diode line). A common 635 nm red diode is the cheap fallback, but Wood 1976's own data shows "a large decrease… between 620 and 640 nm," so expect a weaker response than a true 605-620 nm source. For the 480 nm and 680 nm arms, standard blue (~450-473 nm) and deep-red (~650-685 nm) diodes are cheap and common — no gap issue there.

To get a **consistent, uniform spot covering a well (6-20 mm diameter) over a 0.5-2 ft throw**: collimate the diode, then pass it through a fixed Galilean beam expander (e.g. Thorlabs GBE, 5x on a ~3 mm collimated beam gives ~15 mm output) to hold the spot size steady across that distance range, then an engineered diffuser (e.g. Thorlabs ED1-C series) to flatten the Gaussian profile into a top-hat so every point in the well gets the same fluence rate. An iris/aperture after the diffuser can clip to an exact diameter within the 6-20 mm target if needed. Keep total power modest — Iwatsuki's tested range was ~0.1-1.1 W/m^2, and Kuhnel-Kratz & Hader saw kinesis disappear above ~10 W/m^2 — a few mW, attenuated by driver current or an ND filter, is plenty.

Suggested next-experiment design

1. Primary arm: **610 nm and 620 nm**, measuring photokinesis (swimming velocity), not just avoidance/phototaxis.
2. Secondary arms: **480 nm and 568 nm** as backup candidates.
3. Comparison arm: **680 nm**, kept in the design specifically to directly test Iwatsuki's claim against our own setup, now that we know it's an unreplicated outlier rather than settled fact.
4. Keep fluence rates under ~10 W/m^2 per the Kühnel-Kratz ceiling, and record chamber width/depth so results are comparable across studies.
5. If 610-620 nm produces a clear photokinesis response, that's a novel result (nobody has published a photokinesis measurement at that wavelength) supporting a shared photoreceptor with the phototaxis/photophobic response. If it doesn't, that keeps the second-photoreceptor question open — but 680 nm still wouldn't be confirmed by our data either, so a positive result there would itself be noteworthy given the literature gap.

Source status

Paper	Status
Iwatsuki 1992	Local reference library (not committed — copyright), values verified
Wood 1976	Local reference library (not committed — copyright), values verified
Song 1980	Local reference library (not committed — copyright), values verified
Kuhnel-Kratz & Hader 1994	Local reference library (not committed — copyright), read — no wavelength data
Kim/Prusti/Song/Hader 1984	Local reference library (not committed — copyright, ILL came through), values verified

Full-text PDFs for all five papers above are kept outside this repo (in a local reference library) rather than committed. Wood 1976, Song 1980, Kim/Prusti/Song/Hader 1984, and Kuhnel-Kratz & Hader 1994 came with ILL cover sheets with an explicit "no further copies, electronic or paper" restriction that doesn't permit redistribution. Iwatsuki 1992's PDF has no such cover-sheet notice, but it's still a copyrighted journal article — it was committed to this repo and pushed to the public fork earlier in this project before this concern came up; that commit has since been removed from history and force-pushed, and the PDF moved to the same local reference library as the others. Every value we cite from all five papers has been independently re-verified against the primary-source PDF text before being written into these docs, even though the PDFs themselves live outside the repo. The derived Figure 3 crop image (`references/iwatsuki-1992-figure3-action-spectrum.png`) is still committed, since it's a small excerpt used for commentary/critique rather than the full paper.